Consolidated Experiment Log

Clarifying Prompts for Robust English-Arabic Question-Type Classification

Prompt Engineering Research Project

Mediterranean Institute of Technology (MedTech)

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Fused and reformatted from the detailed chronological experiment log and the final cross-model summary.

April-May 2026

Field	Value
Project	Bilingual prompt classification for six answer-type labels
Course context	Prompt Engineering research project - MedTech
Date covered	April-May 2026
Primary task	Classify English and Arabic questions into NUMBER, LOCATION, PERSON, DESCRIPTION, ENTITY, or ABBREVIATION
Evaluation design	Eight task-instruction prompt variants per question; final label selected by majority vote across the 8 outputs
Metrics	Accuracy, macro-F1, average prompt sensitivity, average consistency, class-level accuracy, and error analysis
Source files fused	Pasted text.txt + final_experiment_log.pdf

Editorial note
Duplicate planning sections were condensed into motivation notes, while completed experiment results were preserved as the authoritative log entries. The final Gemini compact-variant results from the short PDF summary were integrated into the cross-model section.

Contents

1. Executive Summary 4

2. Experimental Setup and Label Schema 4

2.1 Task . . . 4
2.2 Prompt-variant evaluation design . . . 4
2.3 Labels . . . 5
2.4 Final targeted disambiguation rule . . . 5

3. Prompt Progression Summary 5

4. Detailed Experiment Logs 6

4.1 Pilot_01 - Falcon3 tiny pipeline pilot . . . 6
4.2 Pilot_02 - Expanded balanced dataset . . . 6
4.3 Dataset update - Cleaned expanded dataset . . . 7
4.4 Pilot_03 - Cleaned dataset results . . . 7
4.5 Pilot_04 - Definition-based prompts . . . 7
4.6 Pilot_05 - Definition-based few-shot prompts . . . 8
4.7 Pilot_06 - DESCRIPTION vs ENTITY disambiguation . . . 9
4.8 Pilot_07 - Frozen prompt on unseen validation data . . . 9

5. Final Cross-Model Validation 10

5.1 Cross-model performance metrics . . . 10
5.2 Cross-model robustness metrics . . . 10
5.3 Open-model class-level results . . . 11
5.4 ALLaM cross-model log entry . . . 11
5.5 AceGPT cross-model log entry . . . 11

6. Gemini Compact-Variant Closed-Model Experiment 11

7. Generated Files and Reproducibility Notes 12

7.1 Core output families . . . 12
7.2 Final Gemini compact-variant files . . . 12
7.3 Reproducibility controls . . . 13

8. Consolidated Conclusion and Next Steps 13

8.1 Final conclusion . . . 13
8.2 Main limitations . . . 13
8.3 Recommended next steps . . . 13

Source Integration Note 14

1. Executive Summary

This consolidated experiment log documents the full progression of the English-Arabic question-type classification project. The study began with a tiny Falcon3 pilot, expanded to a balanced bilingual dataset, tested prompt modifications step by step, froze the strongest prompt strategy, then validated it on unseen data and across open and closed models.

The final prompt strategy combined explicit label definitions, one few-shot example per label, and a targeted rule separating DESCRIPTION from ENTITY. This progression substantially improved both task performance and robustness, especially in Arabic, where earlier prompts produced more label-boundary confusion.

Each question was evaluated under eight prompt variants, and the final label was selected by majority vote. Therefore, a 120-question experiment produced 960 prompt-level generations: 480 English generations and 480 Arabic generations. This repeated-prompt design was used to measure not only accuracy, but also prompt sensitivity, consistency, and majority-vote stability.

The final unseen-validation run with Falcon3 reached 1.0000 English accuracy and 0.9833 Arabic accuracy. Cross-model validation showed near-perfect performance for Falcon3 and ALLaM, lower but still strong performance for AceGPT, and a specific Gemini compact-variant weakness on Arabic ABBREVIATION.

Experiment	Prompt	N	Runs	Eng. Acc	Ar. Acc	Eng. F1	Ar. F1	Key finding
Pilot_01	Tiny pipeline pilot	12	96	0.8333	0.5000	0.7778	0.4444	Initial debugging run; Arabic much weaker.
Pilot_02	Expanded balanced dataset	120	960	0.8667	0.6833	0.8301	0.6599	ENTITY ambiguity exposed.
Pilot_03	Cleaned dataset + simple prompts	120	960	0.8667	0.6667	0.8303	0.6366	Dataset cleaning alone did not solve errors.
Pilot_04	Label definitions	120	960	0.9500	0.8000	0.9475	0.7830	Definitions improved accuracy and stability.
Pilot_05	Definitions + few-shot examples	120	960	0.9667	0.8667	0.9663	0.8540	Examples strongly reduced Arabic sensitivity.
Pilot_06	Definitions + examples + DESCRIPTION/ENTITY rule	120	960	0.9833	0.9500	0.9833	0.9500	Targeted rule fixed most Arabic DESCRIPTION errors.
Pilot_07	Frozen final prompt on unseen validation	120	960	1.0000	0.9833	1.0000	0.9833	Final prompt generalized to unseen data.

2. Experimental Setup and Label Schema

2.1 Task

The task was single-label question-type classification. For each English or Arabic question, the model predicted the expected answer type from six fixed labels. All experiments used prompt-level outputs and majority voting over eight prompt variants.

2.2 Prompt-variant evaluation design

The eight prompt variants were task-instruction variants rather than subjective politeness levels. They preserved the same six-label schema, the same label definitions, the same few-shot examples when used, the same DESCRIPTION/ENTITY disambiguation rule when used, and the same exact-label-only output constraint. The variants changed instruction wording and compactness while keeping the classification task constant.

Because the variants did not measure politeness, no native-speaker politeness questionnaire was required for this experiment log. For each question, the model produced eight prompt-level outputs, and majority voting selected the final label. This controlled prompt-engineering experiment used the repeated-prompt design as diagnostic evidence for accuracy, prompt sensitivity, consistency, and majority-vote stability.

2.3 Labels

Label	Operational meaning
NUMBER	The expected answer is a number, date, year, quantity, age, measurement, or count.
LOCATION	The expected answer is a place, city, country, region, landmark, or geographical location.
PERSON	The expected answer is a person or named human figure.
DESCRIPTION	The expected answer should define, explain, describe a function, purpose, use, meaning, or process.
ENTITY	The expected answer is a concrete or named non-person, non-location, non-number, non-abbreviation thing, such as an object, animal, planet, language, currency, device, material, software, gas, element, food, sport, or instrument.
ABBREVIATION	The expected answer is the expansion or meaning of an acronym, abbreviation, or shortened form.

2.4 Final targeted disambiguation rule

The decisive prompt addition was a targeted DESCRIPTION vs ENTITY rule. The rule instructed the model to choose DESCRIPTION when the question asks for an explanation, meaning, definition, function, purpose, use, or process; and to choose ENTITY when the question asks for the name of a specific object, animal, planet, language, currency, device, material, gas, software, element, metal, instrument, food, or sport.

Arabic-specific guidance treated forms such as the following as DESCRIPTION when they ask for a definition or explanation:

ما هو ...؟ / ما هي ...؟ / ماذا يعني ...؟ / ما معنى ...؟ / ما وظيفة ...؟ / ما استخدام ...؟

3. Prompt Progression Summary

The Falcon3 prompt progression shows the central experimental narrative. Simple prompts were enough for clear classes such as NUMBER, LOCATION, and PERSON, but weak for semantically broad or ambiguous labels. Definitions improved the results, few-shot examples stabilized Arabic predictions, and the targeted disambiguation rule produced the strongest development-set result.

Pilot	English Sens	Arabic Sens	English Cons	Arabic Cons
Pilot_01	0.0000	0.1810	1.0000	1.0000
Pilot_02	0.0346	0.1685	0.8565	0.6394
Pilot_03	0.0414	0.1588	0.8727	0.6370
Pilot_04	0.0048	0.0825	0.9213	0.7051
Pilot_05	0.0116	0.0137	0.9194	0.8106
Pilot_06	0.0080	0.0057	0.9713	0.8889
Pilot_07	0.0048	0.0105	0.9917	0.9542

Main trend
Arabic accuracy rose from 0.6667 in Pilot_03 to 0.9500 in Pilot_06, while Arabic sensitivity fell from 0.1588 to 0.0057. This supports the interpretation that better prompt design reduced Arabic label-boundary confusion rather than merely improving raw accuracy.

4. Detailed Experiment Logs

4.1 Pilot_01 - Falcon3 tiny pipeline pilot

Field	Value
Date	April 2026
Experiment ID	Pilot_01_Falcon3_7B_English_Arabic_Question_Type_Classification
Model	tiiuae/Falcon3-7B-Instruct
Dataset	12 total questions: 6 English + 6 Arabic; one example per label per language
Prompting	8 prompt templates per language; majority vote over the 8 outputs
Total generations	$12 \times 8 = 96$ prompt-level generations

Language	Accuracy	Macro-F1	Avg Sensitivity	Avg Consistency	Correct
English	0.8333	0.7778	0.0000	1.0000	5/6
Arabic	0.5000	0.4444	0.1810	1.0000	3/6

Interpretation: The pipeline worked end to end, but the dataset was too small for strong conclusions. English predictions were more accurate and stable than Arabic predictions. Arabic errors already showed confusion among DESCRIPTION, ENTITY, LOCATION, and ABBREVIATION.

Generated files: raw_predictions_hf.csv and metrics_hf.csv.

4.2 Pilot_02 - Expanded balanced dataset

Field	Value
Date	May 2026
Experiment ID	Pilot_02_Falcon3_7B_Expanded_English_Arabic_Question_Type_Classification
Model	tiiuae/Falcon3-7B-Instruct
Dataset	120 total questions: 60 English + 60 Arabic; 10 examples per label per language
Prompting	8 task-instruction prompt variants; majority vote across outputs
Total generations	$120 \times 8 = 960$ prompt-level generations (480 English + 480 Arabic)

Language	Accuracy	Macro-F1	Avg Sensitivity	Avg Consistency
English	0.8667	0.8301	0.0346	0.8565
Arabic	0.6833	0.6599	0.1685	0.6394

Language	NUMBER	LOC.	PERSON	DESC.	ENTITY	ABBR.
English	10/10	10/10	10/10	10/10	2/10	10/10
Arabic	10/10	10/10	10/10	4/10	3/10	4/10

Interpretation: The expanded dataset revealed that ENTITY was the main weak class. English ENTITY reached only 2/10, while Arabic DESCRIPTION, ENTITY, and ABBREVIATION were also weak. The results suggested that some errors reflected label ambiguity, not just model failure.

4.3 Dataset update - Cleaned expanded dataset

Before Pilot_03, the expanded dataset was revised while preserving the same balanced structure: 120 total questions, 60 per language, and 10 examples per label per language. The main goal was to reduce ENTITY ambiguity by replacing or clarifying examples that overlapped strongly with LOCATION or ABBREVIATION.

The cleaned ENTITY class focused more clearly on non-person, non-location, non-number, non-description, and non-abbreviation answers such as devices, animals, planets, currencies, software applications, gases, elements, and metals.

4.4 Pilot_03 - Cleaned dataset results

Field	Value
Date	May 2026
Experiment ID	Pilot_03_Cleaned_Dataset
Model	tiiuae/Falcon3-7B-Instruct
Dataset	Cleaned 120-question balanced dataset
Prompting	8 simple label-only task-instruction variants; majority vote across outputs
Total generations	$120 \times 8 = 960$ prompt-level generations (480 English + 480 Arabic)

Language	Accuracy	Macro-F1	Avg Sensitivity	Avg Consistency	Correct
English	0.8667	0.8303	0.0414	0.8727	52/60
Arabic	0.6667	0.6366	0.1588	0.6370	40/60

Language	NUMBER	LOC.	PERSON	DESC.	ENTITY	ABBR.
English	10/10	10/10	10/10	10/10	2/10	10/10
Arabic	10/10	10/10	10/10	4/10	3/10	3/10

Interpretation: Cleaning the dataset did not improve overall accuracy. English accuracy remained 0.8667, and Arabic accuracy decreased slightly to 0.6667. This indicated that the main issue was not only dataset noise; the prompt needed clearer label meanings.

Generated files: raw_predictions_hf_pilot03.csv, metrics_hf_pilot03.csv, sample_level_results_hf_pilot03.csv, error_analysis_hf_pilot03.csv, and prompt_level_errors_hf_pilot03.csv.

4.5 Pilot_04 - Definition-based prompts

Field	Value
Date	May 2026
Experiment ID	Pilot_04_Definition_Prompts
Model	tiiuae/Falcon3-7B-Instruct
Dataset	Same cleaned 120-question dataset as Pilot_03
Prompting	8 prompt variants with explicit definitions for all six labels

Field	Value
Total generations	$120 \times 8 = 960$ prompt-level generations (480 English + 480 Arabic)

Language	Accuracy	Macro-F1	Avg Sensitivity	Avg Consistency	Correct
English	0.9500	0.9475	0.0048	0.9213	57/60
Arabic	0.8000	0.7830	0.0825	0.7051	48/60

Language	NUMBER	LOC.	PERSON	DESC.	ENTITY	ABBR.
English	10/10	10/10	10/10	10/10	7/10	10/10
Arabic	10/10	10/10	8/10	4/10	6/10	10/10

Interpretation: Explicit definitions improved both accuracy and macro-F1 in English and Arabic, and reduced prompt sensitivity. The largest gains appeared in ambiguous categories, especially English ENTITY, Arabic ENTITY, and Arabic ABBREVIATION. Arabic DESCRIPTION remained weak at 4/10.

4.6 Pilot_05 - Definition-based few-shot prompts

Field	Value
Date	May 2026
Experiment ID	Pilot_05_Definitions_FewShot
Model	tiiuae/Falcon3-7B-Instruct
Dataset	Same cleaned 120-question dataset
Prompting	8 prompt variants with label definitions and one example per label
Total generations	$120 \times 8 = 960$ prompt-level generations (480 English + 480 Arabic)

Language	Accuracy	Macro-F1	Avg Sensitivity	Avg Consistency	Correct
English	0.9667	0.9663	0.0116	0.9194	58/60
Arabic	0.8667	0.8540	0.0137	0.8106	52/60

Language	NUMBER	LOC.	PERSON	DESC.	ENTITY	ABBR.
English	10/10	10/10	10/10	10/10	8/10	10/10
Arabic	10/10	10/10	10/10	4/10	8/10	10/10

Interpretation: Few-shot examples improved Arabic performance substantially and reduced Arabic average sensitivity from 0.0825 to 0.0137. The main unresolved issue was Arabic DESCRIPTION, which remained at 4/10 and was often predicted as ENTITY.

Error pattern: English had two majority-vote errors, both ENTITY questions predicted as LOCATION. Arabic had eight majority-vote errors, mainly DESCRIPTION questions predicted as ENTITY.

4.7 Pilot_06 - DESCRIPTION vs ENTITY disambiguation

Field	Value
Date	May 2026
Experiment ID	Pilot_06_Description_Entity_Disambiguation
Model	tiiuae/Falcon3-7B-Instruct
Dataset	Same cleaned 120-question development dataset
Prompting	Definitions + few-shot examples + targeted DESCRIPTION/ENTITY rule
Total generations	$120 \times 8 = 960$ prompt-level generations (480 English + 480 Arabic)

Language	Accuracy	Macro-F1	Avg Sensitivity	Avg Consistency	Correct
English	0.9833	0.9833	0.0080	0.9713	59/60
Arabic	0.9500	0.9500	0.0057	0.8889	57/60

Language	NUMBER	LOC.	PERSON	DESC.	ENTITY	ABBR.
English	10/10	10/10	10/10	10/10	9/10	10/10
Arabic	10/10	10/10	9/10	9/10	9/10	10/10

Interpretation: The targeted disambiguation rule was highly effective. Arabic DESCRIPTION improved from 4/10 in Pilot_05 to 9/10 in Pilot_06, and Arabic ENTITY improved from 8/10 to 9/10. This was the strongest development-set result.

- **Remaining error 1:** English ENTITY question about the main language spoken in Brazil was predicted as LOCATION.
- **Remaining error 2:** Arabic PERSON question about who developed relativity was predicted as DESCRIPTION.
- **Remaining error 3:** Arabic DESCRIPTION question about a black hole was predicted as ENTITY.
- **Remaining error 4:** Arabic ENTITY question about the main language used in Brazil was predicted as LOCATION.

4.8 Pilot_07 - Frozen prompt on unseen validation data

Field	Value
Date	May 2026
Experiment ID	Pilot_07_Unseen_Validation
Model	tiiuae/Falcon3-7B-Instruct
Dataset	New unseen validation dataset: 120 questions, 60 English + 60 Arabic, 10 examples per label per language
Prompting	Exact frozen Pilot_06 prompt strategy reused without modification across 8 variants; majority vote
Total generations	$120 \times 8 = 960$ prompt-level generations (480 English + 480 Arabic)

Language	Accuracy	Macro-F1	Avg Sensitivity	Avg Consistency	Correct
English	1.0000	1.0000	0.0048	0.9917	60/60
Arabic	0.9833	0.9833	0.0105	0.9542	59/60

Language	NUMBER	LOC.	PERSON	DESC.	ENTITY	ABBR.
English	10/10	10/10	10/10	10/10	10/10	10/10
Arabic	10/10	10/10	9/10	10/10	10/10	10/10

Interpretation: The final prompt generalized successfully to new unseen data. The earlier Arabic DESCRIPTION and ENTITY problems were fully resolved in this validation set, with both classes reaching 10/10.

Only one majority-vote error remained. The Arabic question below had gold label PERSON, but the model predicted NUMBER with prediction counts NUMBER = 6 and ENTITY = 2:

من ألّف سوناتا ضوء القمر؟

5. Final Cross-Model Validation

After Pilot_07, the final prompt was frozen and evaluated across Falcon3, ALLaM, AceGPT, and Gemini. The same unseen validation dataset, labels, eight prompt variants, majority-vote method, and evaluation metrics were used. The only changed variable was the model.

Because the frozen prompt was evaluated under eight variants for each of four models, the final unseen cross-model validation produced 3,840 prompt-level generations in total. This total is separate from earlier Falcon3 pilot and development runs.

5.1 Cross-model performance metrics

Model	Type	Eng. Acc	Ar. Acc	Eng. Macro-F1	Ar. Macro-F1
Falcon3-7B-Instruct	Open	1.0000	0.9833	1.0000	0.9833
ALLaM-7B-Instruct-preview	Open	0.9833	1.0000	0.9833	1.0000
AceGPT-v2-8B-Chat	Open	0.9167	0.8833	0.9087	0.8618
Gemini-2.5-Flash compact variants	Closed	0.9833	0.8333	0.9833	0.7778

5.2 Cross-model robustness metrics

Model	English Sens	Arabic Sens	English Cons	Arabic Cons
Falcon3-7B-Instruct	0.0048	0.0105	0.9917	0.9542
ALLaM-7B-Instruct-preview	0.0059	0.0059	0.9833	0.9833
AceGPT-v2-8B-Chat	0.0385	0.0313	0.8787	0.8824
Gemini-2.5-Flash compact variants	0.0253	0.0225	0.9662	0.9704

5.3 Open-model class-level results

Model	Language	NUMBER	LOC.	PERSON	DESC.	ENTITY	ABBR.
Falcon3	English	10/10	10/10	10/10	10/10	10/10	10/10
Falcon3	Arabic	10/10	10/10	9/10	10/10	10/10	10/10
ALLaM	English	10/10	10/10	10/10	10/10	9/10	10/10
ALLaM	Arabic	10/10	10/10	10/10	10/10	10/10	10/10
AceGPT	English	10/10	10/10	10/10	10/10	5/10	10/10
AceGPT	Arabic	10/10	10/10	10/10	10/10	3/10	10/10

Falcon3 and ALLaM achieved near-perfect performance. AceGPT performed well on NUMBER, LOCATION, PERSON, DESCRIPTION, and ABBREVIATION, but struggled with ENTITY. Its errors were concentrated in ENTITY questions that were often predicted as DESCRIPTION, suggesting possible over-application of the disambiguation rule.

5.4 ALLaM cross-model log entry

Field	Value
Model	humain-ai/ALLaM-7B-Instruct-preview
Experiment ID	CrossModel_ALLaM_7B_Pilot07
Prompt	Frozen final prompt: definitions + few-shot examples + DESCRIPTION/ENTITY rule
Dataset	Pilot_07 unseen validation dataset
Main result	English 0.9833 accuracy; Arabic 1.0000 accuracy
Only error	English ENTITY question: ”What natural satellite orbits Earth?” predicted as LOCATION after a 4/4 LOCATION vs ENTITY tie.

5.5 AceGPT cross-model log entry

Field	Value
Model	FreedomIntelligence/AceGPT-v2-8B-Chat
Experiment ID	CrossModel_AceGPT_8B_Pilot07
Prompt	Frozen final prompt: definitions + few-shot examples + DESCRIPTION/ENTITY rule
Dataset	Pilot_07 unseen validation dataset
Main result	English 0.9167 accuracy; Arabic 0.8833 accuracy
Main weakness	ENTITY performance: 5/10 in English and 3/10 in Arabic.

6. Gemini Compact-Variant Closed-Model Experiment

Field	Value
Model	Gemini-2.5-Flash
Experiment type	Closed-model compact variants of the frozen final prompt
Prompting	Eight compact prompt variants; majority voting
Total generations	$120 \times 8 = 960$ closed-model prompt-level generations (480 English + 480 Arabic)
Generation settings	Temperature 0.0, max output tokens 64, seed 42 where supported, thinking budget 0
Result summary	English remained strong; Arabic ABBREVIATION was the main failure mode.

Gemini was not evaluated with a single prompt. It was evaluated on the same 120 validation questions under eight compact prompt variants, producing 960 closed-model generations: 480 English and 480 Arabic. The reported Gemini accuracy and macro-F1 are majority-vote results across the eight compact variants.

Gemini reached 0.9833 English accuracy and 0.8333 Arabic accuracy. English had one majority-vote error: ”What does HTTP stand for?” tied between ABBREVIATION and DESCRIPTION and was resolved as DESCRIPTION. In Arabic, Gemini correctly handled NUMBER, LOCATION, PERSON, DESCRIPTION, and ENTITY, but all ten Arabic ABBREVIATION questions were classified as DESCRIPTION. This lowered Arabic macro-F1 to 0.7778.

Interpretation of Gemini result
The Gemini compact-variant result shows that a model can be stable across prompt variants while still being systematically wrong on a language-specific class. The unresolved issue is not overall Arabic performance; it is Arabic acronym/abbreviation handling under compact prompt variants.

7. Generated Files and Reproducibility Notes

7.1 Core output families

Across the experiments, the pipeline generated the following types of files:

- raw_predictions: every prompt-level output for every sample.
- metrics: aggregate accuracy, macro-F1, sensitivity, and consistency by language.
- sample_level_results: one row per question with majority prediction and correctness.
- class_level_results: per-class results for each language.
- error_analysis: majority-vote errors and their prediction distributions.
- prompt_level_errors: individual prompt-level errors.
- summary_tables: additional summary or bootstrap files when available.

The raw_predictions files are essential because they preserve every prompt-level generation. The sample_level_results files summarize the majority-vote prediction for each question. Keeping both file families makes it possible to audit the repeated-prompt design, inspect tie cases, and separate majority-vote accuracy from prompt-level diagnostic evidence.

7.2 Final Gemini compact-variant files

- results/raw_predictions/raw_predictions_crossmodel_gemini_2_5_flash_compact_variants.csv
- results/sample_level/sample_level_results_crossmodel_gemini_2_5_flash_compact_variants.csv
- results/metrics/metrics_crossmodel_gemini_2_5_flash_compact_variants.csv
- results/class_level/class_level_results_crossmodel_gemini_2_5_flash_compact_variants.csv
- results/error_analysis/error_analysis_crossmodel_gemini_2_5_flash_compact_variants.csv
- results/prompt_level_errors/prompt_level_errors_crossmodel_gemini_2_5_flash_compact_variants.csv
- results/summary_tables/bootstrap_ci_gemini_compact_variants.csv

7.3 Reproducibility controls

- Use the exact frozen final prompt for validation and cross-model runs.
- Keep temperature at 0.0 for deterministic or near-deterministic behavior.
- Record model names and versions exactly as used.
- Preserve raw prompt-level predictions, not only aggregate metrics.
- Report tie cases transparently, especially when majority vote is evenly split.
- Keep API keys out of notebooks, logs, PDFs, and repository files.

8. Consolidated Conclusion and Next Steps

8.1 Final conclusion

The experiment progression supports the conclusion that prompt design strongly affects bilingual English-Arabic question-type classification. Simple prompts produced lower Arabic accuracy and higher sensitivity, especially around DESCRIPTION, ENTITY, and ABBREVIATION. Adding definitions, few-shot examples, and a targeted DESCRIPTION/ENTITY rule improved both accuracy and stability.

The final frozen prompt generalized well to unseen validation data with Falcon3 and transferred strongly to ALLaM. AceGPT showed a model-specific weakness in ENTITY, while Gemini compact variants showed a language-specific weakness in Arabic ABBREVIATION. Therefore, the final prompt strategy is effective in this controlled pilot setting, but model-specific and label-specific behavior remains important.

8.2 Main limitations

- The dataset is small and controlled, so results should be presented as pilot-scale evidence rather than a broad benchmark claim.
- The label schema still contains naturally difficult boundaries, especially ENTITY versus DESCRIPTION and ENTITY versus LOCATION.
- Tie-breaking can affect final majority-vote labels when prompt variants split evenly.
- Gemini compact variants indicate that high consistency does not guarantee correctness for every language-specific class.

8.3 Recommended next steps

- Add a dedicated Arabic ABBREVIATION rule and more Arabic acronym-expansion examples, then rerun Gemini compact variants.
- Increase the dataset size and include more natural questions beyond the controlled examples.
- Define and document tie-breaking behavior explicitly.
- Use the final log as the experiment-history appendix for the research paper or repository documentation.

Source Integration Note

This document fuses the detailed chronological experiment log from Pasted text.txt with the final summary and Gemini compact-variant information from final_experiment_log.pdf. The content was rewritten into a consistent experiment-log format, with duplicate planning sections condensed and final completed results retained.